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$$f(x) = \frac{2x^2 - 4x + 1}{\sqrt{x}}$$

Toepassing quotient regel.

$$\begin{aligned} f'(x) &= \frac{(4x - 4) \cdot \sqrt{x} - (2x^2 - 4x + 1) \cdot \frac{1}{2}x^{-\frac{1}{2}}}{x} \\ &= \frac{4x^{\frac{3}{2}} - 4x^{\frac{1}{2}} - (x^{\frac{3}{2}} - 2x^{\frac{1}{2}} + \frac{1}{2}x^{-\frac{1}{2}})}{x} \\ &= \frac{4x^{\frac{3}{2}} - 4x^{\frac{1}{2}} - x^{\frac{3}{2}} + 2x^{\frac{1}{2}} - \frac{1}{2}x^{-\frac{1}{2}}}{x} \\ &= \frac{3x^{\frac{3}{2}} - 2x^{\frac{1}{2}} - \frac{1}{2}x^{-\frac{1}{2}}}{x} \\ &= 3x^{\frac{1}{2}} - 2x^{-\frac{1}{2}} - \frac{1}{2}x^{-\frac{3}{2}} \end{aligned}$$

References